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Geovisualization
TAA4

Predictive Mapping for Invasive Weed Management

Context:

The state of Idaho has a long history of logging. The state of Idaho was allowed to start commercially cut timber in 1892. From here Idaho timber fueled the needed industry and trade that was booming in post gold rush west. Northern Idaho in particular had the largest piece of the sector due to the fact of the terrain being easier to navigate and, most importantly, the farthest inland port linked to the Pacific Ocean, the Port of Lewiston. This port connected the waterways of the Snake River to the Columbia allowing northern Idaho to have still a thriving timber industry. It was during this time that weeds started to appear in the western United States (Reichard and White). With over 21 million acres of forestland covering 38% of Idaho's total area, the state now conducts approximately 150,000 acres of timber harvest annually across 57 ranger districts. Each harvest operation creates the disturbed conditions that invasive weeds exploit. Invasive weeds thrive when local populations are disturbed or removed leaving no competition. Invasive weeds have traits like rapid reproduction, large numbers of offspring, and short lifespans. These species typically inhabit unstable or unpredictable environments where they can quickly exploit resources and reproduce before conditions change (MacArthur and Wilson). In biology we call these types of species R-selected species (MacArthur and Wilson), there are K-selected invasive species, the opposite of R-selected but they usually not considered threats and are easy to manage. Once invasives are established in an ecosystem they can fundamentally change how the system responds to environmental factors like fire (D'Antonio and Vitousek) and out competing forage that local fauna evolved to eat (Mack et al.).

The economic impact is staggering: invasive weeds cost the United States collectively $2 billion annually (DiTomaso). Once invasives are established they can be nearly impossible to remove but it is possible to manage them. Managing invasive weeds can range from manual removal, chemical, and biological. The management can range from physical removal, pesticides and herbicides and biological, the introduction of the targeted weed natural predator.

This study is to determine what Forest Service Ranger Districts need enhanced weed management for post timber harvests as well as which genus of invasive weeds are the most problematic.

Audience and Persona: Department Heads and District Rangers

James Jones is the head of Idaho's Rangeland management and has a hard time preparing district Rangers on how properly allocate resources. Some districts over buy on herbicide while others, that may need to increase their effects, do not. This represents our primary audience: mid-level range management and district leaders, Rangers. Different weeds may need unique applications to combat. If head of Ranger units were prepared for the statistical chance of certain weeds then they could prepare. For instance some weeds need early chemical treatment but Skeleton Weed (*Chondrilla* sp.) responds well with spotted moth biological treatments (Andreas et al.). Catching these moths and putting them in coolers is a time sensitive operation having certain levels of statistical certainty can convince local Rangers to allocate the man power or funds to acquire these assets for the following growing season post timber harvest.

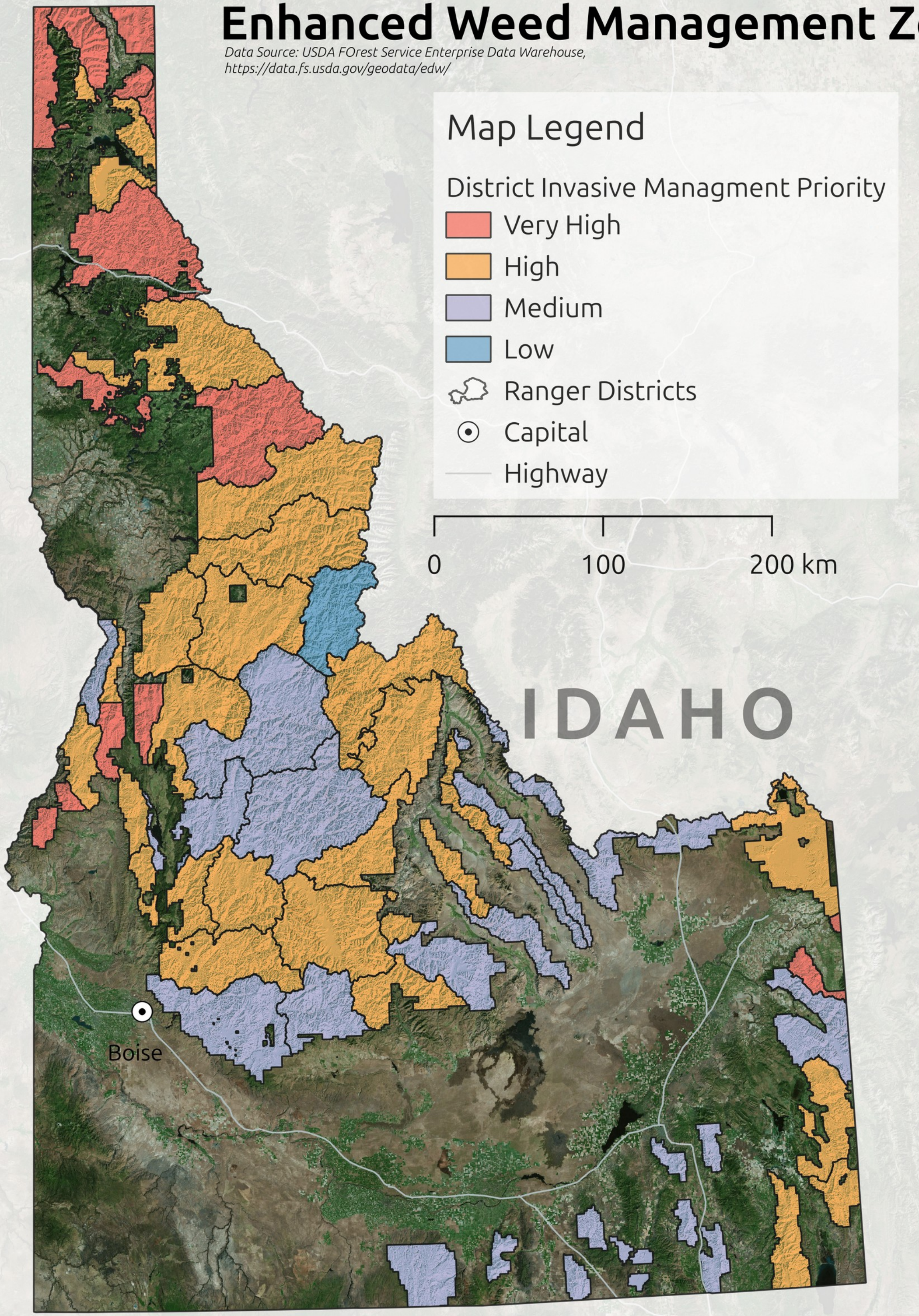
Participatory Map Design: Collaborative Validation and Refinement Network

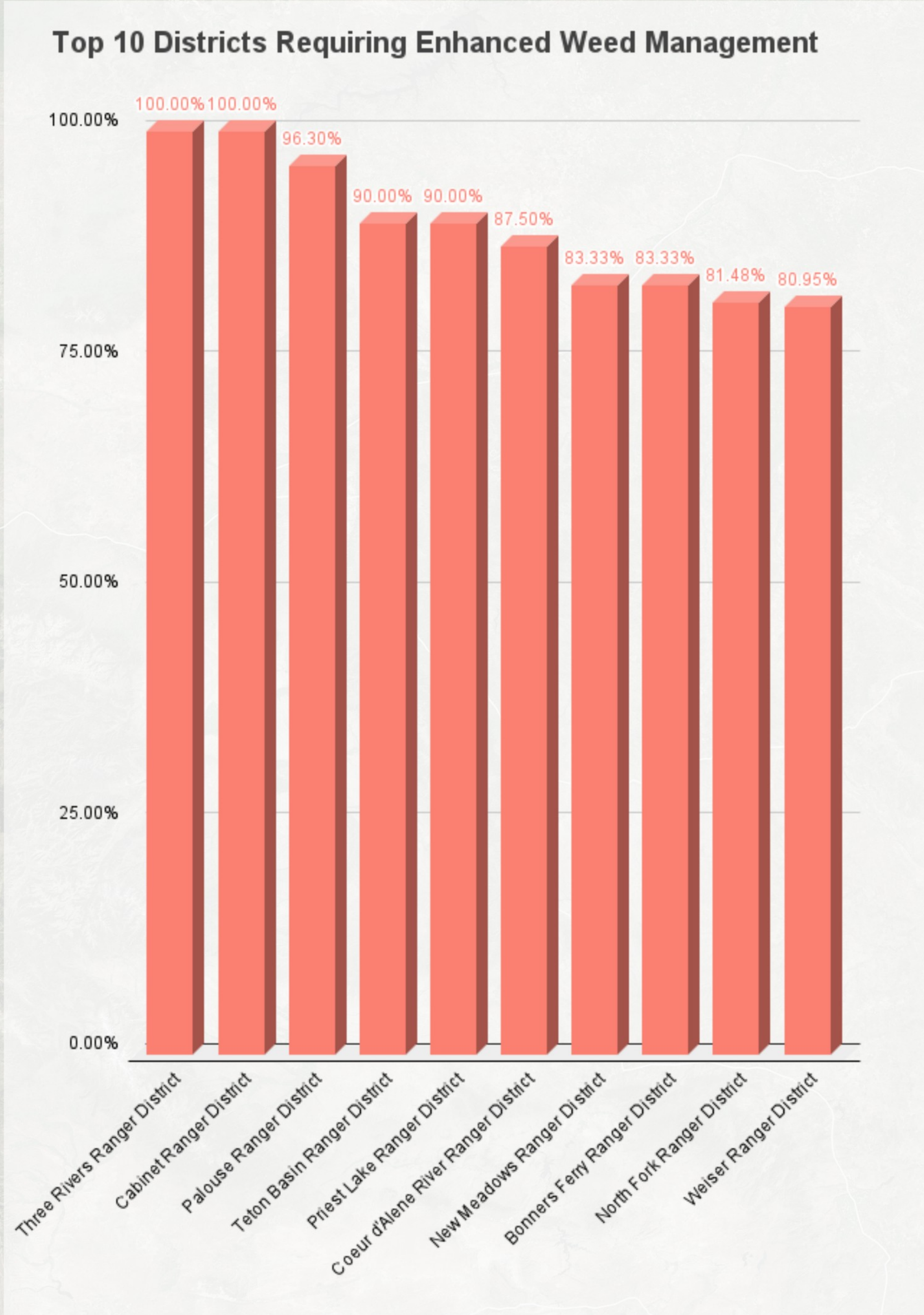
The Forest Service is constantly doing invasive weed surveys alongside roadways, post fires, and post timber harvests. The participatory nature of this project is already rooted in the organization. The key difference is Ranger districts often respond to invasive establishments rather than preventative measures. Building on this existing framework, we propose a structured "Predictive Validation Partnership" that leverages the collaborative nature of the Forest Service with yearly meetings where James and his peers share operational challenges and best practices. Each participating district receives predictive maps showing invasion probability zones for upcoming harvest areas, categorized by elevation and dominant predicted species. James and peers implement control measures in predicted high-risk areas while leaving predicted low-risk areas as controls. This project creates a ton of added value to planning, budget allocation, and anticipating needs.

Main Visualization

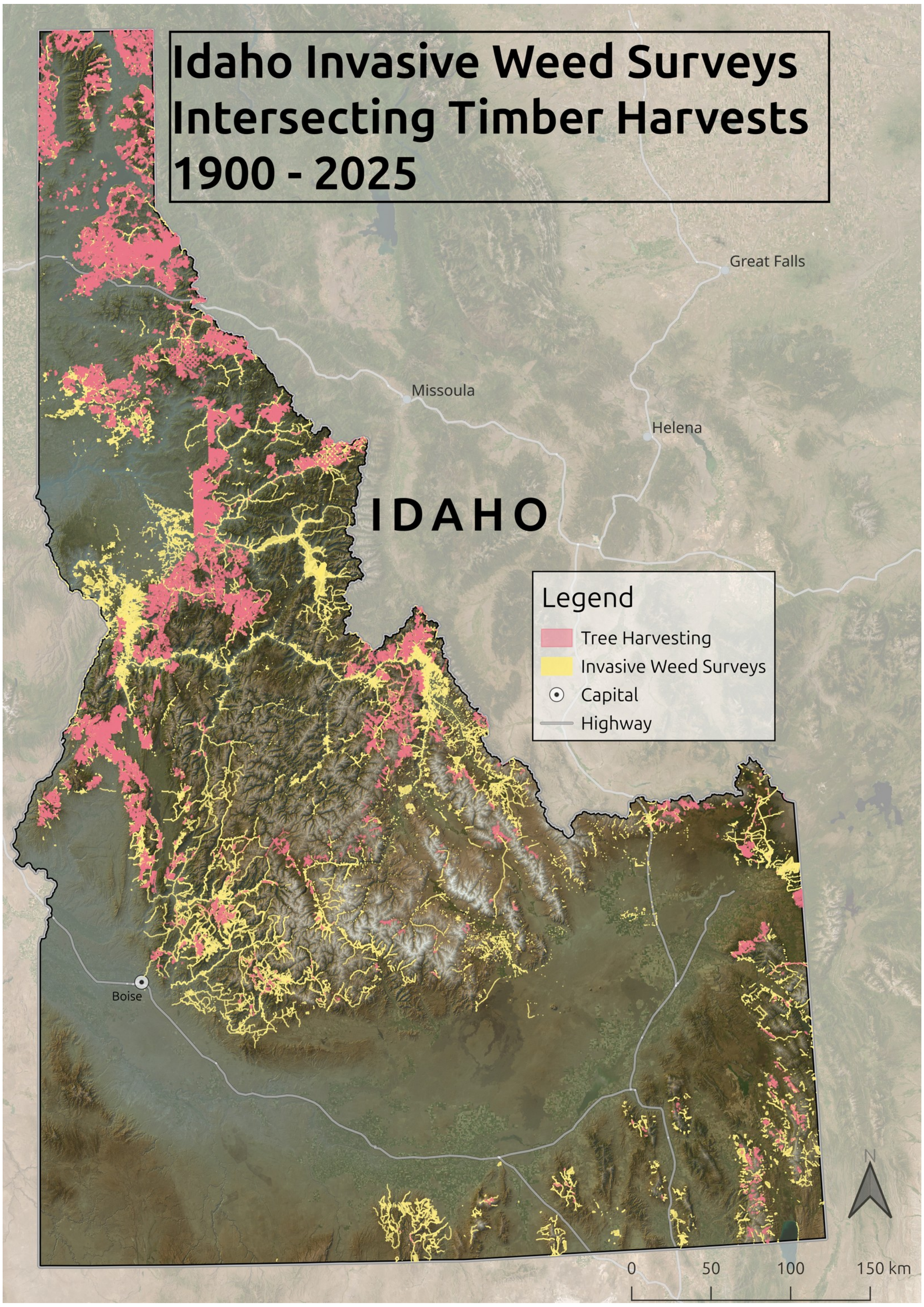
Enhanced Weed Management Zones for Post Timber Harvests in Idaho

Data Source: USDA FOrest Service Enterprise Data Warehouse, https://data.fs.usda.gov/geodata/edw/





Transparency and Data Provenance



Map 2

For this analysis I synthesized 3 datasets from the United States Department of Agriculture Forest Service FSGeodata Clearinghouse and 1 Digital elevation model (DEM) from the United States Geological Survey and 1 deprived from project analysis.

1. Invasive Weed Survey (S_USA.Bio_InvasivePlantCurrent.shp): This is a comprehensive spatial database containing 77,648 polygon surveys from 1934 to 2024. Each polygon includes a range of survey information including taxonomic classification (SCIENTIFIC name), administrative codes (PROJECT_CO, PROTOCOL_N), and infestation metrics (INFESTED_A, COVER_PCT). The polygon size is determined by how much of said invasive weed is seen. (Map 2)
2. Timber Harvested Zones (S_USA.Actv_TimberHarvest.shp): This dataset showed every timber harvest from 1900-2024. Designated timber zone has a harvest rate period around 15-20 years. (Map 2)
3. Forest Service District Boundaries (ForestDistricts.shp): The 57 Ranger Districts throughout the whole state of Idaho. This provided the spatial framework for the invasive management need.
4. Species Management Comparison Data (species_baseline_harvest_comparison.csv): My derived dataset containing 1,07r records that measures the relationship between baseline invasive species occurring in the districts and in harvest zones within those districts. This distinguishes between “harvest-specialist” species that only appear in post timber disturbances and “harvest”-enhanced” species that appear in harvest zones and in areas without harvesting.
5. Digital Elevation Model (IdahoDEM.tif): High-resolution terrain data at 30m provided by the USGS.

From here I began the data transformation and processing. Several major transformations were needed to apply to create an analysis ready project. First I did spatial intersection analysis where a three-way spatial intersection between districts, weeds, and harvest were made to differentiate baseline invasion patterns from post-harvest establishment. This operation created an identification which invasive species capitalized on timber harvesting disturbance, this created the mechanism for the predictive modeling. Second I reprojected all datasets to EPSG:4269 to maintain spatial consistency during analysis. Third I did taxonomic aggregation where full scientific names were parsed to purely genus level. I did this because many of the top species were from the same genus and therefore didn’t show the diversity of genus. Fourth I calculated slope and aspect using Horns algorithm with aspect converted to sine/cosin components to handle the directional nature of the data. Fifth the genus’s were then categorized into harvest specialists, only appearing in disturbed areas, and harvest-enhanced where they were found in harvest zones and outside harvest zones. My final step was data quality enhancements through Density-Based Spatial Clustering of Applications with Noise (DBSCAN) where outlier detection removed irrelevant elevation values.

Three-Step Analytical Framework

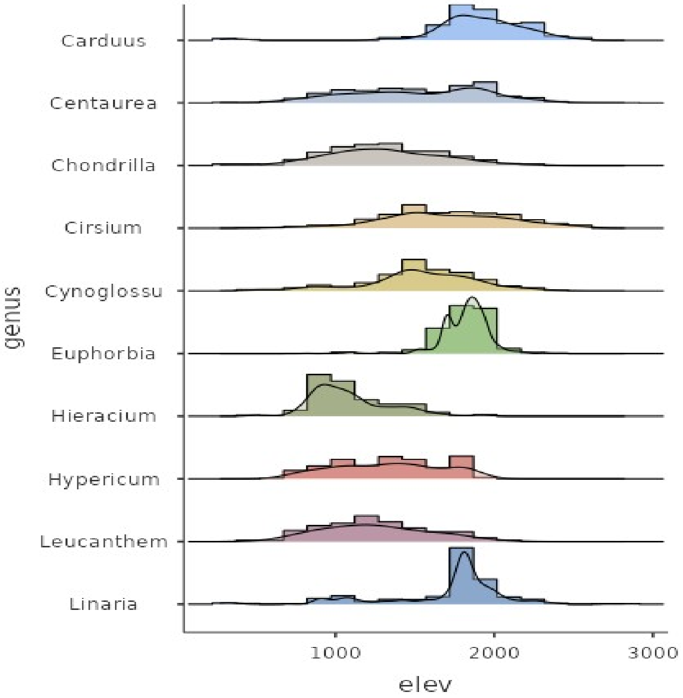


Figure 1

Individual Attribute Analysis: For individual attribute analysis many key insights were derived from this analysis. 88.5% of sampled locations require more weed management, this is reflecting systematic invasion pressure in and around post harvest environments. Harvest-specialist species (e.g., Linaria, Chondrilla) represent novel management challenges requiring proactive interventioConcentration around mid-elevation zones (1,200-2000m) (Figure 1) this aligns with ease of access to the timber as well as more mild climates.

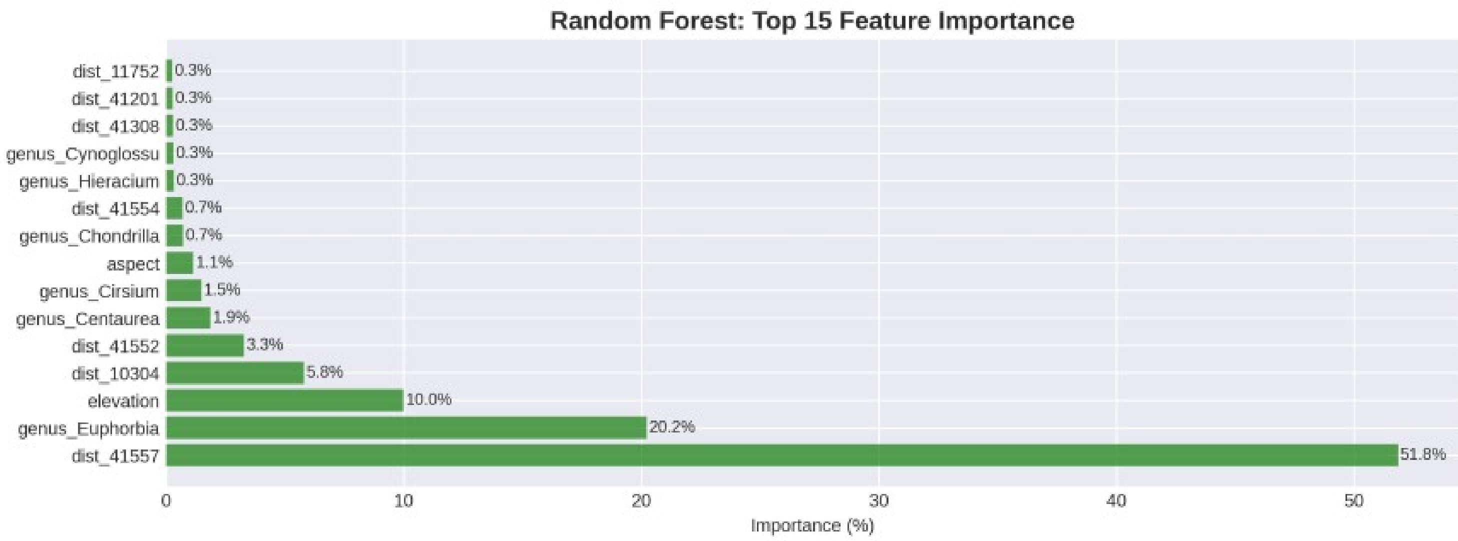


Figure 2

Cross-Attribute Analysis: Here I preformed Random Forest Classification model (500 trees, max depth=10). This achieved a crass validated AUC: 0.987 ± 0.006 with minimal over-fitting (training-test gap: -0.002) (Figure 2). District 41557 (Westside Ranger District) having 51.8% indicates localized combinations of harvest intensity, invasion history, and terrain. The second largest feature is the plant genus Euphorbia, this genus showed a 3.2 higher establishment rates in recently harvested areas compared to non harvested areas. The third major feature is elevation. Random forests (RF) provides very high classification accuracy, a method of determining variable importance, making it particularly effective for invasive plant species classification (Cutler, D. Richard)

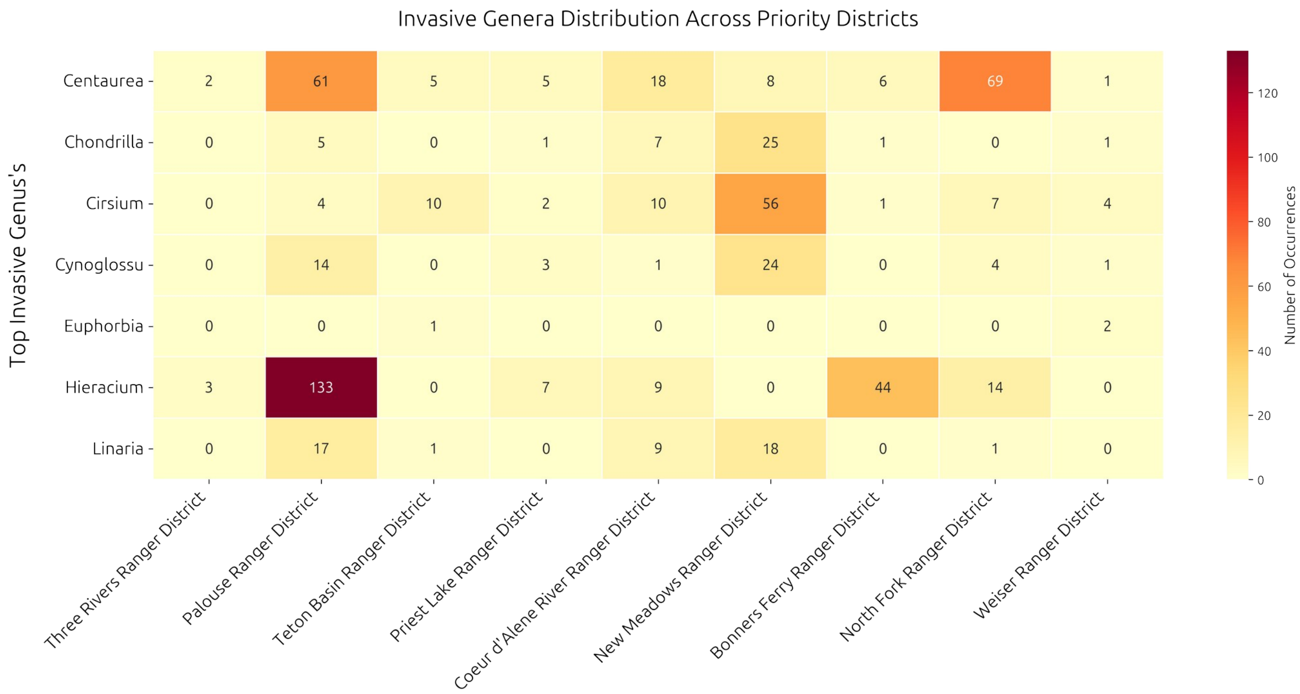


Figure 3

This analysis also created one of the most important results. Which top genus's are found in the ranger districts. These invasives are found in and outside of the harvest polygons with large counts. Showing non local prevalence but district wide.

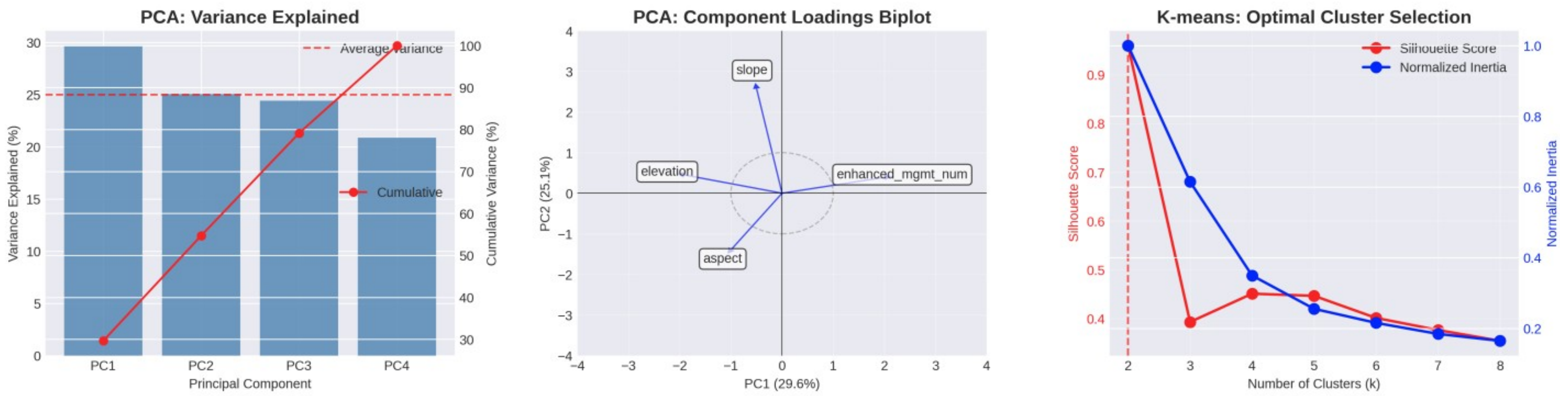


Figure 4

Variable Reduction: This final step I did Principal Component Analysis (PCA) on environmental and management variables revealing the underlying structure of invasion risks (Figure 3). PC1 represents a harvest management gradient with disturbed low elevation sites requiring intensive management attention. This aligns with Idahos r-selected invasive species thriving in warmer environments. The Clustering Analysis multiple algorithms converge on a two cluster solution distinguishing harvest impacted from the rest of the undisturbed areas in the district.

K-means optimal k=2 with a silhouette score: 0.960. DBSCAN has two distinct clusters separating harvest with 1,847 points from no harvested areas. Lastly a Gaussian Mixture Model with clear bimodal distribution with 94% assignment certainty.

The integration of timber harvest data creates fascinating insights towards understanding the dynamics of invasive species distribution. The distinction between harvest specialists and harvest enhanced species enables targeted pre-deployment of management resources. This allows districts to plan if they need biological controls, chemical, or manual before a harvest takes place. . Having district specific risk profiles (51.8% predictive importance) combined with harvest future harvest schedules gives vital tools to district Rangers. This project reaffirms known research that timber harvest creates disturbed conditions that invasive weeds thrive in (Rejmanek, Marcel, and Mark A. Davis).

Reflection

In designing the final layout I made a deliberate choice to focus on a single operational variable, the absolute number of weed infestations that demand enhanced post-harvest attention in each Idaho ranger district, because intensity of invasive weeds are the metric that ultimately determines budgets, crew deployment, and herbicide ordering not only the presence. That variable is expressed on the map as a simple four-class, single-hue sequential choropleth running from butter-yellow (Low) through pumpkin (Medium) to deep brick red (Very High). The ramp is perceptually ordered, CVD-safe, and re-uses the warm accent that punctuated. I made Idaho on the left side of the layout so it is firstly perceived with the districts, while the accompanying bar chart answers how intense and which districts first. To avoid duplicating information, the bars rank only the districts requiring enhanced weed management that are “Very High” on the map. I also made sure the bars are the same hue as the Very High.

I leaned heavily on Gestalt heuristics to usher the viewer toward a correct reading with minimal cognitive load. Similarity binds polygons to bars through their shared hue family and ranking. Proximity and a pale-gray common region panel pack the map and chart into one perceived unit. The districts are de saturated with a soft-blurring with the underlying hill shade so thematic colors “float” above terrain. I wanted to share a 3D effect with the graph, as well as, giving the Idaho map more dimensions. Hierarchy is clear but not abrasive: darkest reds and the tallest bars carry the greatest visual weight, while subtitle, legend, and peripherals have a soft opacity around 70%. Color decisions further lighten cognitive load. The warm sequential ramp follows Brewer guidelines (Brewer, Cynthia A., et al.) for ordered data and ensures equal perceived steps. Spatial context is preserved by subtle terrain shading, yet relief is dimmed to 30 % opacity so it supports, rather than competes with, the thematic layer, an application of Ware’s guidance on luminance contrast (Ware, Colin.). Typography unifies the composition: Ubuntu Sans is used at constant sizes, satisfying the law of common fate by letting text blocks move together across scales. All map elements required by basic cartographic convention, title, legend, scale, north arrow, data source, are present, but each is purposefully understated and logically placed to balance the page: legend beneath the title to avoid covering northern “red zones,” scale centered under the legend to allow Idaho to fit as much as the page as possible, north arrow tucked lower-right to offset the heavy chart entire right. I intentionally omitted district labels to prevent overwhelming the reader with text. For the primary audience, ranger-district staff, the shapes alone uniquely identify their own jurisdictions, so adding names would contribute little new information while masking the color pattern that conveys priority. I’m still not confident if that was the right choice, in the future I would revisit the decision by using IDs or even an interactive dashboard, however, suppressing labels keeps the visual hierarchy clean and lets the choropleth signal stand out.

Together these design choices let the stake holders answer a very important question, which district needs the most management need, in under five seconds. With the provided visuals and systematic analyses this project produced a coherent, low-friction information experience.

Works Cited

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